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MILITARY SPECIFICATION

NICKEL BAR, FLAT WIRE (RIBBON) AND STRIP (FOR ELECTRICAL USE)

1. SCOPE

1.1 This specification covers one grade of nickel for electronic use in the forms of bar, flat wire (ribbon) and strip (see 6.1).

1.2 Classification.

1.2.1 Grade and tempers. The one grade of nickel shall be commercially pure. It shall be furnished in the following tempers, as specified (see 6.2).

Annealed
As drawn

1.2.2 Finish. Unless otherwise specified in the contract or order, nickel bar, flat wire (ribbon) and strip shall be furnished in bright finish.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Test Method Std. No.151 – Metals; Test Methods

FSC 9525, 9530,9535

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MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 – Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E 8 - Tension Testing of Metallic Materials
E 18 – Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

E 92 – Vickers Hardness of Metallic Materials
E 140- Standard Hardness Conversion Tables for Metals

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Chemical requirements.

3.1.1 The material shall conform to the chemical requirements shown in table I.

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Table I. Chemical composition

Element	Percent[1]
Nickel[2]	99.0 min.
Others, total	1.00
Iron	0.30
Copper	.20
Manganese	.35
Carbon	.15
Silicon	.20
Sulfur	.008

[1]Maximum unless otherwise indicated.
[2]Cobalt counting as nickel

3.1.2 Analysis shall be made routinely only for the elements specifically mentioned in table I. If, however, the presence of other elements is suspected, or indicated in amounts greater than the specified limits further analysis shall be made to determine that the total of other elements is not in excess of these limits.

3.1.2.1 The contractor shall furnish analysis of each melt showing the percentage of each of the elements designated in table I.

3.1.2.2 Chemical analysis by the contractor of the individual melts may be waived at the discretion of the procuring activity provided that the producer's method of the composition control is acceptable or that all material in the lot can be identified as being from melts previously analyzed and found to be in conformance with the chemical composition requirements.

3.2 Mechanical properties. The mechanical properties of nickel bar, flat wire (ribbon) and strip shall be in accordance with the requirements specified in tables II and III.

Table II. Tension test requirements

	Size	Temper	Tensile strength, psi min.	Tensile strength, psi max
Bar	All sizes	Annealed	55,000	-
Cold drawn or cold rolled		As drawn	65,000	-
Flat wire (ribbon)	All sizes	Annealed	-	75,000
		As drawn	80,000	-

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Table III. Hardness of cold-rolled strip[1]

Thickness (All widths)	Hardness Rockwell B (max.) or equivalent[2]
0.005 to 0.010, incl.	70
Over 0.010 to 0.024, incl.	68
Over 0.024 to 0.125, incl.	64

[1]Deep-drawing quality, annealed temper
[2]For hardness conversions, see table IV of ASTM Standard Hardness Conversion Tables for Metals, E 140. Also see 6.3

3.3 Dimensional tolerances, bar.

3.3.1 Distance between parallel surfaces. Bar, measured on the distance between parallel faces, shall not vary at any point from the specified dimensions by more than the amounts shown in table IV.

Table IV. Tolerances on distance between parallel faces of bar

Distance between parallel faces, inches	Tolerance, inch (minus only)
1/16 to 3/16, excl.	0.002
3/16 to 15/16, incl.	.003
Over 15/16 to 1-1/2, incl.	.004
Over 1-1/2 to 2, incl.	.005
Over 2 to 2-3/4, incl.	.006
Over 2-3/4 to 4, incl.	.007

3.3.2 Length

3.3.2.1 Nominal lengths. Nominal lengths between 6 and 14 feet may be specified. The minimum length shall be the specified length. The maximum length shall be the specified length plus 2 feet, 1/2 inch.

3.3.2.2 Random lengths. All random lengths shall be between 4 and 16 feet, with not more than 25 percent by weight under 9 feet.

3.3.2.3 Cut straight lengths. If cut straight lengths are specified, the lengths shall be within the limits specified in table V.

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Table V. Tolerances of cut straight lengths

Length, feet	Tolerance, inch (plus only)
Up to 6, incl.	0.125
Over 6	.250

3.3.3 Straightness. The deviation from straightness shall not be greater than the amounts in table VI.

Table VI. Straightness tolerances of bar

Distance between parallel faces, inch	Permissible deviation (depth of arc) inch
All sizes	0.025 per foot of length

3.4 Dimension tolerances, flat wire (ribbon).

3.4.1 Thickness. The thickness of the flat wire (ribbon) shall not vary by more than the amounts shown in table VII.

Table VII. Thickness tolerances for flat wire (ribbon)

Thickness, inch	Tolerance, inch (plus and minus)
Up to 0.006, incl.	0.0003
Over 0.006 to 0.009, incl.	.0005
Over 0.009 to 0.025, incl.	.0007
Over 0.025 to 0.048, incl.	.001
Over 0.048 to 0.187, incl.	.002

3.4.2 Width. The width of flat wire (ribbon) shall not vary by more than the amounts shown in table VIII.

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Table VIII. Width tolerances for flat wire (ribbon)

Width, inch	Tolerance, inch (plus and minus)
Up to 0.031, incl.	0.005
Over 0.031 to 0.061, incl.	.008
Over 0.061 to 0.099, incl.	.010
Over 0.099 to 0.187, incl.	.016

3.4.3 Length. If cut straight lengths are specified, the lengths shall be within the limits as specified in table V.

3.4.4 Permanent set. The flat wire (ribbon) shall be sufficiently rigid so that when 1-1/4 turns are cut free from a spool, the diameter of the resulting loop shall not exceed the flange diameter of the spool on which the wire is furnished.

3.5 Strip.

3.5.1 Thickness. The thickness of strip shall not vary by more than the amounts shown in table IX.

Table IX. Thickness tolerances for strip[1]

Thickness, inch	Tolerance (all widths), inch (plus and minus)
Under 0.006,	10[2]
0.006 to 0.015, incl.	0.001
Over 0.015 to 0.050, incl.	.0015
Over 0.050 to 0.093, incl.	.0025
Over 0.093 to 0.125, incl.	.004

[1]Measured 3/8 inch or more from edge except for strip under 1 inch in width which is measured at any place.
[2]Expressed as percent of nominal thickness.

3.5.2 Width. The width of strip shall not vary by more than the amounts shown in table X.

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Table X. Width tolerances for strip[1]

Thickness, inch	Width, Inches	Tolerance, inch (plus and minus)
0.009 and under	All widths	0.005
Over 0.009 to 0.024, incl.	All widths	.007
Over 0.024 to 0.075, incl.	14 and under	.007
Over 0.075 to 0.100, incl.	14 and under	.009
Over 0.100 to 0.125, incl.	14 and under	.012

[1]For strip with slit edges only.

3.5.3 Length. Strip may be ordered to cut lengths in which case a variation of 0.125 inch over the specified length shall be permitted.

3.5.4 Straightness. The edgewise curvature (depth of arc) of strip shall not exceed 0.05 inch multiplied by the length in feet. Straightness for coiled strip shall be subject to agreement between the manufacturer and contracting officer.

3.6 Workmanship. Material shall be commercially free from dirt, oxide, scale, streaks, stains, grease, oil, lubricants, scratches, pit marks, burns, chattermarks, dents, blisters, laps, grooves, slivers, seams, cracks, inclusions and other injurious imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order, a lot shall consist of material of the same temper, finish, melt, or thickness made by the same manufacturing process, submitted for inspection at one time.

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4.3 Sampling. Samples taken for the purpose of tests prescribed in this specification shall be selected in a manner as to represent correctly the material furnished and avoid needless destruction of finished material when samples representative of the material are available from other sources.

4.3.1 Sampling for chemical analysis. A sample for chemical analysis shall be obtained from each 500 pounds of material that can be identified by melt. Where material cannot be identified by melt, 2 samples shall be obtained. The samples shall be selected in a manner calculated to disclose any nonuniformity of material within the lot. From each sample, not less than 2 ounces of clean millings, drillings, or clippings shall be obtained.

4.3.2 Sampling for mechanical properties.

4.3.2.1 Tension test. Unless otherwise specified in the contract or order, one tension test specimen shall be taken from each 500 pounds of material in the lot.

4.3.2.2 Hardness test. Unless otherwise specified in the contract or order, one test specimen shall be taken from each 500 pounds of material in the lot.

4.3.3 Sampling for visual and dimensional examination. From each lot of material a representative sample shall be selected in accordance with MIL-STD-105, inspection level III, acceptable quality level (AQL) 1.5 percent defective. Pieces selected for dimensional examination may be the same as those selected for visual examination.

4.3.3.1 When flat wire (ribbon) is furnished in coils or on reels or spools, the sample for examination shall be taken from within 10 feet of the outer end. If the sample selected is rejected due to handling marks, and additional 20 feet shall be used for reexamination.

4.4 Examination.

4.4.1 Visual and dimensional. Pieces selected in accordance with 4.3.3 shall be visually examined to determine compliance with the requirements for workmanship (see 3.6) and shall be measured for compliance with the dimensional requirements (see 3.3., 3.4, 3.5) of this specification.

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4.4.2 Straightness. Straightness shall be determined by placing the piece or sample unit on a level surface. Place a straight edge one foot in length parallel to the length of the sample so that it touches the edge being tested and extends an equal distance on both sides of the point of apparent maximum deviation. Measure with a steel scale graduated to 1/100 of an inch the perpendicular distance from the straight edge to the edge of the test sample at the point of maximum deviation from straightness.

4.4.3 Preparation for shipment. Examination of the packaging and packing, and marking for shipment shall be made for conformance with the requirement of section 5.

4.5 Tests.

4.5.1 Tension test specimens.

4.5.1.1 Bar 1/2 inch or less in distance between parallel faces. Bar 1/2 inch or less in distance between parallel faces shall be tested in full size using a free length between the jaws of the testing machine.

4.5.1.2 Bar over 1/2 inch in distance between parallel faces. Test specimens shall be in accordance with method ASTM E 8.

4.5.1.3 Flat wire (ribbon) 1/8 inch thick and over. Specimens shall be in accordance with method ASTM E 8. They shall be machined to size by milling only.

4.5.1.4 Flat wire (ribbon) under 1/8 inch thick. The specimen shall consist of a free length of stock in width received, having square parallel edges free from burrs without machining or stoning.

4.5.2 Hardness test specimens.

4.5.2.1 Strip. Test specimens shall be in accordance with either ASTM E 18 or E 92, whichever is applicable (see 6.3).

4.5.3 Test procedures.

4.5.3.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.1. A single analysis of a composite sample may be made. In case of dispute, analysis by method 111 shall be the basis for acceptance.

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4.5.3.2 Tension tests. Specimens from samples selected in accordance with 4.3.2.1 shall be tested in accordance with ASTM E 8.

4.5.3.3 Hardness tests. Specimens from samples selected in accordance with 4.3.2.2 shall be tested in accordance with ASTM E 18 or E 92. (See 6.3).

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.3 for that sample size, the entire lot shall be rejected subject to the provisions of the section on “Acceptance and Rejection” of MIL-STD-105.

4.6.2 Test features. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5 subject to the provisions of the section on “Rejection and Retests” of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DEVIVERY

5.1 Packaging, level C.

5.1.1 Flat wire (ribbon) and strip.

5.1.1.1 Unless cut lengths are specified, flat wire (ribbon) and strip shall be coiled, spooled or wound on reels in on continuous length as specified in 5.1.1.2, in such a manner that it can be unwound fully without breakage or damage. The gross weight of each coil, reel, or spool shall not exceed 50 pounds for material greater than 0.025 inch in thickness or 25 pounds for material 0.025 inch or less in thickness.

5.1.1.2 Flat wire (ribbon) and strip material less than 3/16 inch in width shall be wound on spools. Material 3/16 inch and wider shall be furnished on cardboard reels or in coils.

5.1.1.3 Spools shall have a minimum hole diameter of 3/4 inch. All flat wire (ribbon) and strip on spools, cardboard reels or in coils shall be wound tape fashion (with no transverse) so that each turn is directly on top of the preceding one. The end of the material shall be attached securely to prevent unwinding during transit.

5.2 Packing, level C. The products shall be separated by size, composition, and temper and packed in accordance with the manufacturer’s standard practice into containers of a type and size commonly used for the purpose, in such a manner a to insure acceptance by carrier for transportation at the lowest rate application and to afford maximum protection from normal hazards of transportation.

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5.3 Marking.

5.3.1 Military. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Bar, flat wire (ribbon), and strip are intended for use as components of magnetrons.

6.1.1 Conservation. Due to the high nickel content of this alloy, its use should be limited to those applications where the use of less critical materials would seriously impair military characteristics.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Form, size, temper and surface finish of material required (see 1.2).
(c) The lengths required and whether “cut to length”, “random mill” lengths, “nominal (stock)” lengths, or multiples are required (see 3.3.2, 3.4.3, 3.5.3).
(d) When tolerances other than those shown are required (see 3.3, 3.4, 3.5).
(e) The thickness of flat wire (ribbon) or strip in decimals of an inch (see 3.4.1 and 3.5.1).

6.3 Hardness testing precautions. Care should be observed in using the Rockwell test on this material, as the results may be affected by the specimen thickness. For thickness under 0.050 inch, the use of the Rockwell superficial or the Vickers hardness test is suggested. For thicknesses of 0.005 to 0.015 inch, incl., the Vickers hardness test should be used

Custodian:
Army - MR

Preparing activity:
Army – MR

Review interest:
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